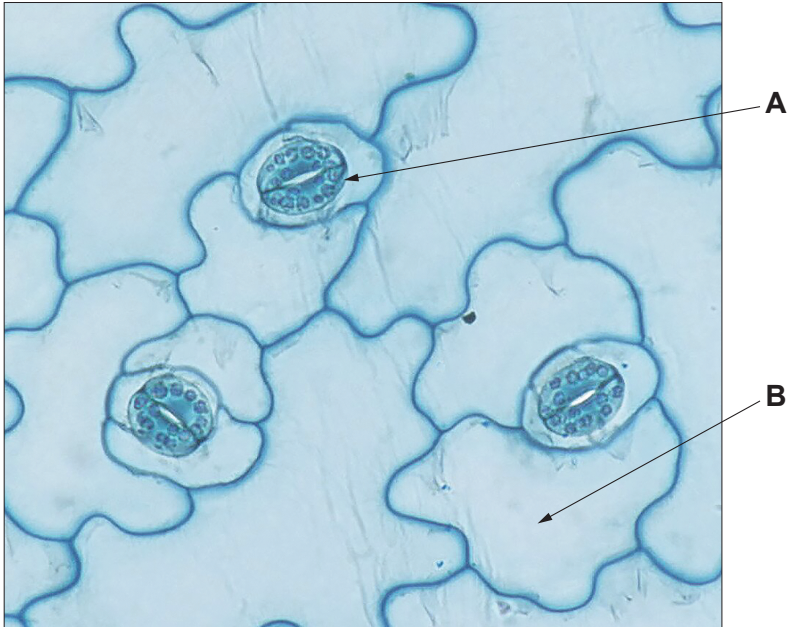


3. **Image 3.1** shows the surface view of stomata found on the lower surface of *Kalanchoe* (*Kalanchoe* sp.) leaves. *Kalanchoe* is a xerophyte.

Image 3.1



(a) (i) Identify the cells labelled **A** and **B** in **Image 3.1**. [1]

A

B

(ii) Describe the mechanism that causes stomata to open. [4]

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(iii) Suggest an advantage to *Kalanchoe* of being able to close stomata. [1]

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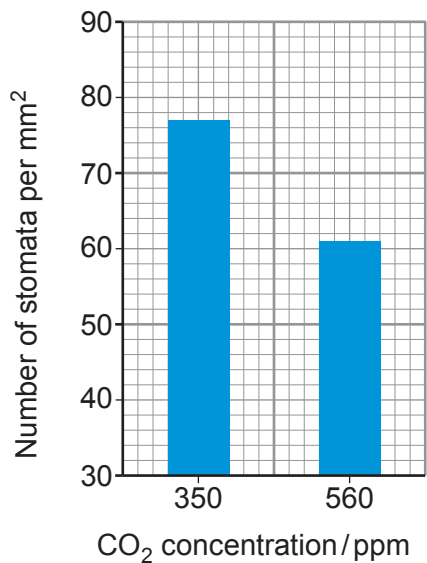
.....



- (b) In the 1990s, researchers investigated the effect of CO₂ concentration on stomatal density in a tree species known as *Ginkgo biloba* using the following method:
- One group of 40 young trees were grown in a greenhouse at atmospheric CO₂ concentrations (350 ppm).
 - A second group of 40 young trees were grown in a different greenhouse at a higher CO₂ concentration (560 ppm).
 - All other conditions were constant for the two groups.
 - After three years growth, the mean number of stomata per mm² was calculated for both groups of trees.

The results are shown in **Graph 3.2**.

Graph 3.2



- (i) State **two** variables that should have been controlled to maintain constant conditions. [1]
- I.
- II.

- (ii) State **one** conclusion that can be drawn from this investigation. [1]
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(iii) Using your knowledge of the functions of stomata, explain the results of this investigation. [3]

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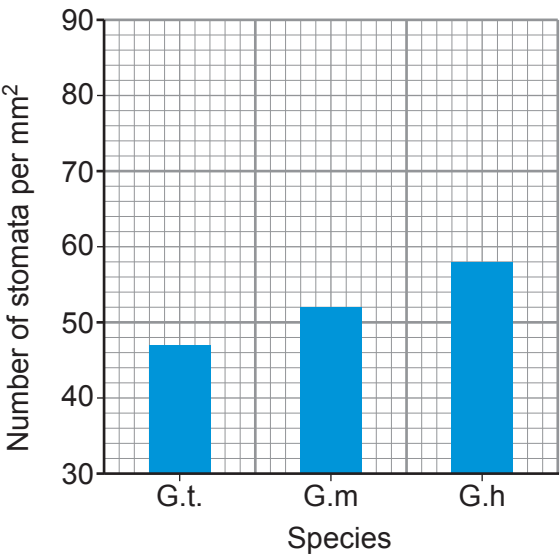
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(c) Darwin described *Ginkgo biloba* as a ‘living fossil’, as fossils of the genus *Ginkgo* have been found in rocks from the Mesozoic era (252 to 66 million years ago). Stomata can be seen on the surface of fossilised leaves. **Graph 3.3** shows the number of stomata per mm² from three different species of the genus *Ginkgo*.

Graph 3.3



Key:

Species	Abbreviation	Age of rocks / million years
<i>G. troedsssonni</i>	G.t	237 – 200
<i>G. marginatus</i>	G.m	200 – 174
<i>G. huttonii</i>	G.h	174 – 163

Using the information in **Graphs 3.2** and **3.3**, suggest what conclusions could be made about the concentration of CO₂ during the Mesozoic era. [2]

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